

Calibrating Geometric Consistency for Reliable 3D Scene Graph Relations

Anonymous submission

Abstract

3D Scene Graphs represent objects and relations in a form useful for spatial reasoning, but relation predictions can remain unreliable even when predicted predicates appear semantically plausible. We study a relation-level failure mode: semantic relation confidence is not necessarily calibrated to the 3D geometry of the same object pair, so a predictor can rank physically inconsistent relations highly. To expose and reduce this failure, we introduce a GeoCalib framework for geometry-checkable relation families. The framework standardizes prediction rows across relation sources, preserves subject-object identity, joins explicit 3D evidence, estimates $p^{\text{geom_valid}}$, and reports calibrated geometry risk under a recall-violation protocol. Two soft fusion instantiations combine semantic and geometry evidence: a calibrated product and a scale-robust rank-average; pooled calibration, Reciprocal Rank Fusion, and hard rules serve as ablation, strong comparison, and diagnostic. On the full official validation split, the calibrated product improves VL-SAT $R@50/R@100$ from 0.9272/0.9635 to 0.9288/0.9683 while reducing Violation@100 from 0.0476 to 0.0333. On the main open-vocabulary relation-source case study, the Open3DSG full-validation recovery branch, the product improves $R@100$ from 0.5161 to 0.6047 while reducing Violation@100 from 0.1242 to 0.0341 under explicit selected-checkpoint and recovery-policy caveats. On a prospectively frozen SGFN target, both the product and pre-specified rank-average pass the joint $R@100/\text{verifier-Violation@100}$ gate, showing that the aggregate effect is not tied to one fusion formula. GT positive/counterfactual checks separate valid and invalid geometry with AUROC/AUPRC of 0.9772/0.9729, and controls rule out geometry-only, distance-only, shuffled-geometry, and wrong-pair explanations. These results support a scoped relation-reliability claim for measured geometry-checkable 3D Scene Graph relations, not formula dominance or broad open-vocabulary graph generation.

Introduction

3D Scene Graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be reliable in the physical scene, not only plausible at the category or language level. A graph edge such as `standing on`, `next`

`to`, or `higher than` is useful only if it describes the actual subject-object pair in 3D.

We study a failure mode in which semantic relation predictors rank plausible relation labels that are geometrically inconsistent for the actual object pair. A relation can sound plausible for two object categories while being contradicted by their contact evidence, distance, or vertical arrangement in the reconstructed scene. This mismatch matters more as 3D Scene Graphs become interfaces for language-facing, open-vocabulary, and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

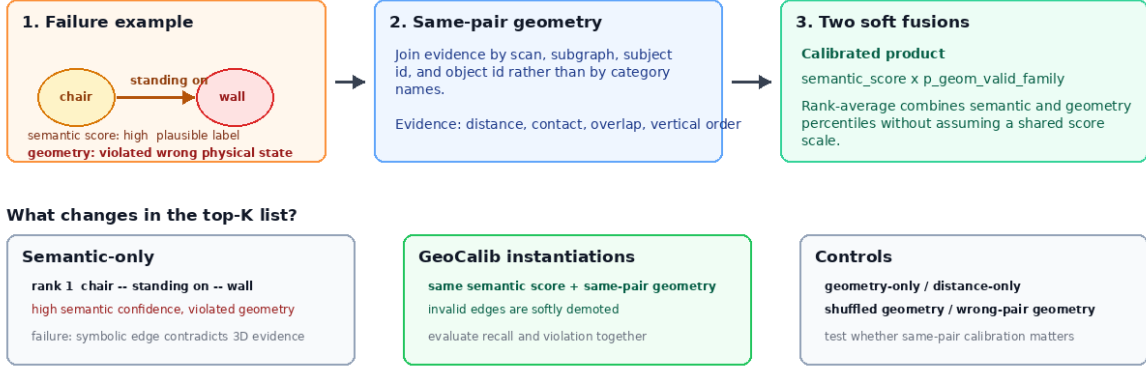
The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: semantic relation confidence is not necessarily a calibrated estimate of relation-level physical consistency. A high relation score can behave as a semantic plausibility score without indicating whether the same subject-object pair satisfies the relation in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-facing and embodied systems can inherit a confident symbolic relation that contradicts the underlying world state.

This motivates a GeoCalib framework. The framework treats an existing predictor as a relation source, standardizes its outputs into identity-preserving prediction rows, joins object-pair geometry evidence by subject and object instance identifiers, estimates calibrated geometry-validity risk, and evaluates relation rankings with both exact-label recall and geometric violation. GeoCalib separates the framework from any one fusion formula. A calibrated product preserves semantic-score magnitudes while penalizing family-conditioned geometry risk; a pre-specified rank-average combines semantic and geometry percentiles without assuming shared source scales. Hard physical rules diagnose strict consistency, pooled calibration is an ablation, and Reciprocal Rank Fusion is a strong comparison. Figure 1 summarizes this path.

We evaluate the framework on geometry-checkable relation families: `support_contact`, `proximity`, and `relative_vertical`. These families are selected be-

GeoCalib: calibrating geometric consistency for relation rows

A semantically plausible edge is not necessarily reliable for the same 3D object pair.



Paper-facing names: product + rank-average = soft instantiations; pooled = ablation; strict rule = diagnostic.

Caption guard: scoped relation-reliability framework, not a broad open-vocabulary generation method.

Figure 1: **GeoCalib in one relation-row example.** A semantically plausible relation can be physically inconsistent for the same object pair. GeoCalib preserves row identity, joins pair geometry, estimates family-conditioned geometry validity, and supports calibrated-product and scale-robust rank-average re-ranking while reporting recall and violation together.

cause their validity can be tested from reconstructed object-pair geometry, they cover contact, metric proximity, and vertical ordering, and they appear in both relation sources with a fixed exact-label denominator. Open3DSG is the main open-vocabulary relation source case study, and VL-SAT is a controlled reproduced closed-set anchor within the same geometry-checkable family scope (Wang et al. 2023; Koch et al. 2024b). Controls test whether the effect can be explained by geometry-only ranking, a generic distance heuristic, shuffled geometry, or wrong-pair geometry. GT-positive and counterfactual checks test the geometry-validity signal independently from prediction re-ranking, while qualitative failure analysis shows how semantic plausibility diverges from physical consistency.

Our contributions are threefold:

1. We identify and formalize a relation-level reliability failure in 3D Scene Graph prediction: semantic relation confidence can rank plausible predicates that are physically inconsistent because it is not calibrated to object-pair geometry.
2. We introduce a GeoCalib framework that standardizes prediction rows, joins identity-preserving 3D evidence, estimates calibrated geometry validity, and exposes calibrated-product and scale-robust rank-average soft fusion instantiations while retaining pooled, RRF, and rule-verified comparisons.
3. We define a recall-violation evaluation protocol for geometry-checkable relation families, including exact-label $R@K$, Violation@K, GT-positive/counterfactual verifier evaluation, and geometry identity controls.

We validate these contributions with Docker-reproducible

evidence on VL-SAT, Open3DSG, and a prospectively frozen SGFN target, including nontriviality controls, family decomposition, failure analysis, and explicit reproduction caveats. The claim is intentionally scoped: GeoCalib measures and improves relation reliability for geometry-checkable 3DSSG families. It does not claim general open-vocabulary 3D Scene Graph generation improvement, arbitrary-source generality, formula dominance, or guaranteed physical correctness.

Related Work

3D Scene Graph relation prediction. 3D Scene Graphs organize indoor scenes as object-relation structures grounded in 3D space (Armeni et al. 2019). The 3RScan/3DSSG setting established predicate recall for 3D relation prediction (Wald et al. 2020), while incremental and online systems study how scene graphs can be constructed from RGB-D streams or spatial perception stacks (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set predicates using edge reasoning, point-cloud features, multimodal cues, spatial knowledge, self-supervised pretraining, object-centric features, and edge-centric relational reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al. 2025; Ma et al. 2026). These methods already use geometry; GeoCalib instead targets the post-source reliability gap that a high semantic relation score need not be calibrated to physical validity for the same object pair.

Open-vocabulary 3D graphs. Open-vocabulary 3D perception provides queryable object and region features (Peng et al. 2023; Takmaz et al. 2023); graph systems then connect

such features to compact scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph work extends this line toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges are language-facing scene representations, plausible text labels must still refer to object pairs that satisfy the corresponding spatial relation.

Other recent systems further connect VLMs, zero-shot graph construction, graph-mediated LLM reasoning, retrieval augmentation, and hallucination mitigation (Saxena and Chiun 2025; Madhavaram et al. 2026; Yu et al. 2025; Zemskova and Yudin 2025; Ogunleye, Abdelrahman, and Lourentzou 2026). We cite them as trend and boundary evidence, not direct GeoCalib baselines.

Geometry-aware relation evidence. A broad set of 3D reasoning methods already uses geometry: edge features, support/contact patterns, affordance geometry, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Huang et al. 2025; Shao et al. 2025; Xie, Pagani, and Stricker 2024). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

RelWitness uses visual-geometric witnesses, calibrated witness quality, and witness-consistent decoding for open-vocabulary 3D scene graph generation (Nguyen et al. 2026). Statistical confidence rescoring refines 3D scene graph predictions from multi-view images using semantic masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). GeoCalib therefore does not claim novelty from visual-geometric evidence, calibration, or rescoring alone. It studies a different post-source contract: identity-preserving geometry joins, recall/violation operating points, and controls over existing relation-source outputs.

Reliability and calibration. Standard R@K asks whether the correct predicate appears near the top, not whether the predicted relation is physically valid in the reconstructed scene. Confidence calibration asks whether model scores reflect empirical correctness likelihood (Guo et al. 2017), and selective prediction makes risk and coverage explicit (Geifman and El-Yaniv 2017). GeoCalib instantiates this reliability view for geometry-checkable 3DSSG families by standardizing prediction rows, joining same-pair 3D evidence, estimating $p^{\text{geom.valid}}$, and reporting recall with violation.

Problem Formulation

Let a relation source produce a set of candidate relation rows for a 3D scene. Each row is written as

$$r_i = (\text{scan}_i, \text{subgraph}_i, s_i, o_i, p_i, a_i, \text{score}_i^{\text{sem}}), \quad (1)$$

where s_i and o_i are subject and object instance identifiers, p_i is the predicate label, a_i is the mapped predicate fam-

ily, and $\text{score}_i^{\text{sem}}$ is the source semantic confidence. The tuple $(\text{scan}_i, \text{subgraph}_i, s_i, o_i)$ is the identity key used to preserve the object pair across prediction, geometry, and ground-truth joins.

For each prediction row, we join object-pair geometry evidence

$$g_i = G(\text{scan}_i, \text{subgraph}_i, s_i, o_i). \quad (2)$$

The evidence can include OBB features, point/local contact evidence, distance, vertical ordering, support/contact subtype information, and missing-evidence reason codes. GeoCalib evaluates only geometry-checkable relation families:

$$\mathcal{A} = \{\text{support_contact}, \text{proximity}, \text{relative_vertical}\}. \quad (3)$$

Rows outside these families are preserved in exports but excluded from the measured violation claim.

The geometry verifier assigns a row-level status $v_i \in \{\text{satisfied}, \text{uncertain}, \text{violated}\}$, and the calibrator estimates pooled and family-conditioned geometry-validity probabilities:

$$p_i^{\text{geom.valid}} = P(v_i \neq \text{violated} \mid g_i, a_i). \quad (4)$$

The calibrated-product instantiation uses the family-conditioned version of this probability, denoted $p^{\text{geom.valid}, \text{family}}$, while the pooled $p^{\text{geom.valid}}$ score remains a calibrated ablation. A second soft instantiation combines within-subgraph semantic and family-geometry ranks. These scores are reliability signals, not proofs of correctness; neither fusion form is assumed universally dominant. We therefore report both soft instantiations together with pooled, rule-verified, RRF, and control variants.

The main recall metric is exact predicate-label R@K over the fixed in-scope ground-truth denominator. Family grouping defines the geometry-checkable reliability scope and violation analysis; it does not relax predicate-label matching. Violation@K is the fraction of top- K in-scope predicted rows whose joined geometry is judged violated. Top- K is formed per evaluated subgraph/source ranking after the measured-family filter; satisfied, uncertain, and violated rows form the violation denominator, while unsupported or missing-join rows are excluded from that denominator and reported through coverage caveats. Only rows with verification status `violated` count as violations. We report recall retention and violation reduction together so that violation reduction cannot be interpreted without its recall tradeoff.

Method

GeoCalib is a framework for 3D scene graph relation predictions. It is not a replacement relation predictor and not a standalone hand-coded verifier. The geometric checks define the measurable validity target for families with observable physical constraints; the method contribution is the identity-preserving row contract, calibrated $p^{\text{geom.valid}}$ signal, operating-point analysis, and controls that test whether geometry is used as a relation-level reliability signal rather than as a generic filter. The framework is built around four

design requirements: preserve relation-row identity, join explicit object-pair geometry, calibrate geometry validity, and evaluate recall and violation jointly. Figure 1 presents these requirements as a pipeline from relation-source outputs to calibrated operating points.

Relation-Row Standardization

Different relation sources expose predictions in different formats. We convert each source output into a shared row contract:

$$(\text{scan}, \text{subgraph}, s, o, p, a, \text{score}^{\text{sem}}, \text{source}). \quad (5)$$

This row is the unit of comparison across VL-SAT, Open3DSG, controls, and future optional sources. The standardized row keeps the semantic score from the source model but adds the fields needed for identity-preserving geometry joins and denominator-stable evaluation.

Identity-Preserving Geometry Join

Geometry is joined by $(\text{scan}, \text{subgraph}, s, o)$, not by category names or unordered object-pair statistics. This matters because the same predicate can be plausible for many object categories while being false for a particular instance pair. The wrong-pair and shuffled-geometry controls directly test this design choice by breaking or perturbing object-pair identity while preserving parts of the geometry distribution.

Family-Specific Geometry Checks

GeoCalib uses relation-family-specific checks for `support_contact`, `proximity`, and `relative_vertical`. For `support/contact`, the evidence includes point/local contact, vertical support, OBB overlap, and support subtypes. For `proximity`, the evidence includes object distance and pair scale. For `relative vertical` relations, the evidence includes centroid and extent ordering along the vertical axis. The verifier returns violated only when the evidence is strong enough; otherwise ambiguous or weak geometry can be marked uncertain. We do not apply this protocol to predicates whose correctness depends primarily on function, social context, affordance, or language pragmatics, because those predicates do not have a stable geometric validity test from the available reconstruction alone.

Calibrated Geometry Fusion Instantiations

The first soft instantiation uses a frozen relation-family-conditioned calibrator:

$$p_i^{\text{geom.valid,family}} = C_{a_i}(\phi(g_i)), \quad (6)$$

where $\phi(g_i)$ is the geometry feature vector and C_{a_i} is the calibrator for the mapped relation family. Support/contact, proximity, and vertical relations need not share one pooled geometry-risk surface. The pooled calibrator,

$$p_i^{\text{geom.valid}} = C_{\text{pooled}}(\phi(g_i)), \quad (7)$$

is retained as the `pooled_risk` ablation. Both calibrators are trained on train-dev calibration positives and counterfactual negatives. The predicate-family map, hard-rule

thresholds, counterfactual construction, and calibrator files are fixed from train-dev calibration artifacts before held-out source-result reporting; held-out prediction rows are not used to retune them.

We first use geometry validity as a soft risk term. For the calibrated product, $R_i^{\text{geom}} = -\log p_i^{\text{geom.valid,family}}$ and

$$U_\lambda(i) = \log \text{score}_i^{\text{sem}} - \lambda R_i^{\text{geom}}. \quad (8)$$

The reported product fixes $\lambda = 1$, giving

$$\text{score}_i^{\text{product}} = \text{score}_i^{\text{sem}} \cdot p_i^{\text{geom.valid,family}}. \quad (9)$$

This preserves the source ranking signal while continuously penalizing calibrated inconsistency; λ is not tuned on held-out source results.

The second soft instantiation removes source-scale assumptions. Within each subgraph, let q_i^{sem} and q_i^{geom} be descending percentile ranks of $\text{score}_i^{\text{sem}}$ and $p_i^{\text{geom.valid,family}}$. Rank-average assigns

$$\text{score}_i^{\text{rank}} = \frac{1}{2}(q_i^{\text{sem}} + q_i^{\text{geom}}). \quad (10)$$

The equal weight is parameter-free and fixed before SGFN inference. The product uses calibrated probability magnitudes; rank-average is scale-robust. We treat them as two GeoCalib fusion instantiations rather than claiming one formula is universally optimal. We additionally report `probabilistic_recalibrated` as the pooled calibrated-risk ablation $\text{score}_i^{\text{sem}} \cdot p_i^{\text{geom.valid}}$, fixed-constant Reciprocal Rank Fusion as a strong comparison, and `rule_verified_point_subtype` as a hard zero-violation diagnostic.

Controls

The control conditions test whether the result can be explained by simpler signals. Geometry-only ranking uses $p_i^{\text{geom.valid}}$ alone and is therefore distinct from the pooled calibrated-risk ablation, which still multiplies semantic confidence by $p_i^{\text{geom.valid}}$. Distance-only ranking tests whether a generic spatial heuristic explains the result. Shuffled-geometry and wrong-pair-geometry controls test whether relation-level object-pair identity is necessary. If these controls perform worse than the calibrated setting, the result supports the framework design rather than a generic “add geometry” explanation.

Experimental Setup

This section fixes scope before results because GeoCalib’s claim depends on denominator transparency as much as metric values. Open3DSG is the main open-vocabulary relation-source case study, while VL-SAT is a controlled reproduced closed-set anchor. Families and denominators are fixed before source-result reporting, yielding a predeclared physically checkable slice: 157 scans, 548 contexts, 36,808 directed pairs, 957,008 VL-SAT prediction rows, 11,254 ground-truth rows, and 3,972 in-scope GT relations. Table 1 records the denominator and source outputs used by Table 2.

Open3DSG is reported as a Docker-reproduced source-output reliability case study, not an official leaderboard reproduction. The selected checkpoint is the official non-averaged BLIP route `epoch=13-step=13104.ckpt`,

Table 1: **Shared GeoCalib evaluation scope.** Recall is exact predicate-label recall over the 3,972-row in-scope GT denominator; family grouping is used for reliability and violation reporting, not for relaxing predicate matches.

Item	Count	Role in claim
Validation scans	157	Shared scope; SGFN’s official test list matches it exactly.
Subgraphs / contexts	548	Object-pair identity denominator.
Directed object pairs	36,808	Candidate relation-row universe.
Ground-truth rows	11,254	Full validation relation annotation rows.
In-scope GT rows	3,972	Exact-label denominator for the three measured relation families.
VL-SAT prediction rows	957,008	Controlled reproduced closed-set anchor output.
Open3DSG prediction rows	695,916	Main open-vocabulary source output.
SGFN prediction rows	957,008	Prospectively frozen exact-label confirmation output.
Geometry-scored rows	160,596	Open3DSG geometry-checkable rows joined to calibrated evidence.

chosen by train-dev validation loss before held-out source-result inspection. Its runtime uses filtered train/dev subgraphs, and the main full-validation branch reaches 548/548 contexts with `OPEN3DSG_MIN_VISIBLE_OBJECTS=2` plus relaxed view regeneration for two scans. The 533/548 covered branch and historical old 377/388 vs R2 388/388 branch remain sensitivity evidence only.

Condition names are fixed before reporting. The calibrated product family_conditional_risk ranks by $\text{score}^{\text{sem}} \cdot p^{\text{geom_valid, family}}$; rank-average combines within-subgraph semantic and family-geometry percentiles with fixed equal weight. `probabilistic_recalibrated` is the pooled calibrated-risk ablation $\text{score}^{\text{sem}} \cdot p^{\text{geom_valid}}$. RRF is a fixed strong comparison, while geometry-only controls rank by $p^{\text{geom_valid}}$ without semantic confidence.

SGFN `full_1160` is the prospective exact-label target. Its source test list exactly matches the frozen 157-scan H001 scope. The target, $K=100$, families, denominator, calibrated product, rank-average, RRF, and bootstrap gates were fixed before correct-checkpoint download and inference. Two pre-inference errata corrected the split identity and an initially mistaken 120 checkpoint URL; all versions remain in provenance. SGFN covers all 36,808 nonself directed pairs. Eleven self-supported by GT rows have no model edge, receive no synthesized prediction, and remain in the 3,972-row recall denominator.

Source boundaries are enforced in prose rather than as a main result table. Open3DSG supports the open-vocabulary relation-source case study; VL-SAT supports the controlled-anchor comparison. FROSS remains runtime-blocked, Qwen-VL is an optional third-source extension, and broad open-vocabulary 3DSSG generation improvement is not claimed without the same Docker, denominator, metric, and audit treatment. All paper-body experiment results are generated through the Docker experiment root. Host-only outputs are not promoted to paper evidence.

Results and Discussion

Table 2 asks whether calibrated geometry consistency reduces violations without collapsing recall. The Open3DSG source exposes the failure most clearly: semantic-only ranking places many geometrically inconsistent edges at the top of the list, especially for the low- K diagnostic. GeoCalib changes that operating point rather than merely pruning

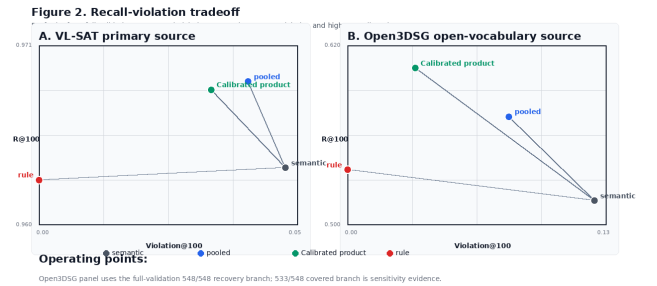


Figure 2: **Recall-violation tradeoff.** Lower Violation@100 and higher R@100 are better. Open3DSG is reported on the full-validation recovery branch; 533/548 and historical 377/388 vs R2 388/388 branches are sensitivity evidence only. The plotted soft operating point is the calibrated product; Table 4 compares fusion forms.

edges. The calibrated product at $K = 100$ improves recall by 8.86 percentage points over the semantic source and reduces violation by 9.01 points; at $K = 5$ and $K = 10$, it also raises recall while moving violation from the high-error semantic regime to the calibrated-risk regime. Pooled risk reduces violations but is weaker than the product on this tradeoff, and the strict-rule row is a zero-violation diagnostic rather than the default method.

VL-SAT provides the controlled anchor. Its semantic ranking is already close to ceiling, so the expected room for recall improvement is small. GeoCalib still reduces violation consistently across the reported K grid and gives a positive R@100 shift. The pooled ablation slightly favors recall at the largest K but leaves more violation, so we report it as a recall-favoring ablation rather than a soft framework instantiation.

A Docker subgraph bootstrap with 1,000 resamples is used as an internal stability check for the point estimates in Table 2. It supports the same paper-facing reading: Open3DSG shows a consistent positive recall shift and negative violation shift across the reported K grid, while VL-SAT shows smaller but stable violation reductions because its source ranking is already near ceiling. We do not print bootstrap ranges in the main table because the comparison is a fixed-source re-ranking analysis, not a repeated-training

Table 2: **Main source results.** Rows use the full official 3,972-row three-family denominator. Open3DSG uses the selected non-averaged checkpoint on the 548/548 recovery branch; 533/548 and historical 377/388 vs R2 388/388 branches are sensitivity evidence only. We report the fixed low-K grid $K \in \{5, 10, 20, 50, 100\}$; $K = 1$ is excluded as a noisy sanity check. Residual calibration risk is reported in the failure analysis rather than hidden in the table. Product is the calibrated-product instantiation; pooled risk is the pooled probabilistic recalibrated ablation.

Source	Condition	R@K					Violation@K				
		@5	@10	@20	@50	@100	@5	@10	@20	@50	@100
Open3DSG	Semantic	0.0368	0.1002	0.1991	0.4096	0.5161	0.5131	0.3255	0.2088	0.1386	0.1242
Open3DSG	Pooled risk	0.0826	0.1581	0.2603	0.3975	0.5723	0.0628	0.0699	0.0654	0.0606	0.0811
Open3DSG	Strict rule	0.0707	0.1314	0.2422	0.4295	0.5368	0.0000	0.0000	0.0000	0.0000	0.0000
Open3DSG	Product	0.0984	0.1921	0.3291	0.4658	0.6047	0.0420	0.0482	0.0441	0.0286	0.0341
VL-SAT	Semantic	0.4194	0.6322	0.8074	0.9272	0.9635	0.0029	0.0082	0.0142	0.0268	0.0476
VL-SAT	Pooled risk	0.4154	0.6322	0.8107	0.9305	0.9688	0.0015	0.0071	0.0120	0.0229	0.0404
VL-SAT	Strict rule	0.4197	0.6317	0.8074	0.9257	0.9627	0.0000	0.0000	0.0000	0.0000	0.0000
VL-SAT	Product	0.4162	0.6309	0.8087	0.9288	0.9683	0.0011	0.0051	0.0109	0.0206	0.0333

Table 3: **Controls and diagnostics.** Geometry-only ranks by $p^{\text{geom.valid}}$ without semantic confidence; pooled risk keeps semantic confidence and uses the pooled calibrator. Shuffled and wrong-pair controls break identity-preserving geometry evidence.

Source	Condition	R@100	Violation@100	Interpretation
Open3DSG	Geometry-only	0.5116	0.0865	Geometry alone does not recover the main tradeoff.
Open3DSG	Distance-only	0.5038	0.1071	Generic distance is weaker than calibrated geometry risk.
Open3DSG	Shuffled geometry	0.2543	0.1998	Breaking row-level geometry destroys reliability.
Open3DSG	Wrong-pair geometry	0.2331	0.1985	Same-pair identity is necessary.
VL-SAT	Geometry-only	0.5184	0.0711	Geometry alone is not a useful ranking source.
VL-SAT	Distance-only	0.5554	0.0981	Distance is a poor proxy for relation reliability.
VL-SAT	Shuffled geometry	0.9494	0.0588	Violations increase when geometry identity is broken.
VL-SAT	Wrong-pair geometry	0.9529	0.0601	Wrong object-pair evidence is worse than GeoCalib.

benchmark, and the 3DSSG literature typically reports point estimates for top-K relation metrics.

Retrospective comparisons separate the framework claim from formula selection: product is the lower-V choice on Open3DSG, whereas rank-based fusion is stronger at VL-SAT $K = 100$. Thus no fusion universally dominates. On the prospectively frozen SGFN target in Table 4, both pre-specified GeoCalib instantiations satisfy the joint gate. Product changes R@100 from 0.9235 to 0.9416 (paired delta 0.0181, 95% CI [0.0134, 0.0233]) and verifier Violation@100 from 0.0630 to 0.0381 (delta -0.0249 , CI $[-0.0270, -0.0229]$). Rank-average reaches 0.9476/0.0277; relative to product its recall-delta CI crosses zero while its V-delta CI is strictly negative. This supports incorporating calibrated same-pair geometry across more than one fusion form, not product superiority.

The aggregate effect is not family-uniform. On SGFN, product support_contact verifier V increases by 0.00450 (95% CI [0.00370, 0.00532]), proximity is unchanged, and relative_vertical supplies most of the reduction. The same support/contact direction appears in the retrospective sources. We therefore claim aggregate scoped reliability and expose the family decomposition rather than claiming every-family improvement.

Table 3 rules out simpler alternatives. Geometry-only is not the same as pooled calibrated re-ranking: it removes se-

mantic confidence entirely. Distance-only is weaker, and shuffled/wrong-pair controls show that object-pair identity is necessary. Thus GeoCalib is not merely ranking by geometry, distance, or hard rules; calibrated same-pair evidence matters.

The GT-based verifier check tests the geometry signal independently: GT positives remain nonviolated at 0.9965, counterfactual negatives are nonsatisfied at 0.9673, and $p^{\text{geom.valid}}$ separates 7,944 rows with AUROC/AUPRC 0.9772/0.9729 and Brier 0.0543. This distinguishes the calibrated reliability signal from a purely hand-written validity script.

Structured audit and visual sanity checks support failure interpretation but are not large-scale human evaluation: a 250-row structured audit reaches quality-issue precision 0.8933, and a 50-row visual spot-check has target-bucket quality-issue rate 0.9333 with contradiction rate 0.0333.

Together, the metrics and controls support a calibrated reliability-layer interpretation, not a new relation generator or broad open-vocabulary SOTA claim. Open3DSG comparisons share the same selected checkpoint, row contract, full-validation denominator, and recovered source payload. The historical R2 branch stays outside Table 2: it covers 388/388 contexts and changes old 377/388 R@100 by only about +0.28 points, so the old missing-context caveat did not create the trend. Its clean-return raw files are row-equivalent

Table 4: **Prospective framework confirmation at $K = 100$.** Product and rank-average are GeoCalib soft instantiations; fixed-constant RRF is a strong comparison. SGFN was prospectively frozen and uses the unchanged 3,972-row denominator. V is verifier-derived, not independent human validity.

Source	Condition	R@100	Violation@100	Interpretation
SGFN	Semantic	0.9235	0.0630	Fresh exact-label source baseline.
SGFN	Product	0.9416	0.0381	Frozen joint gate pass vs semantic.
SGFN	Rank-average	0.9476	0.0277	Frozen joint gate pass vs product.
SGFN	RRF	0.9192	0.0284	Lower V but fails the recall guardrail.

Figure 3: Geometry-backed failure mechanism examples

Same locked Open3DSG qualitative rows, now drawn from preprocessed object point clouds.

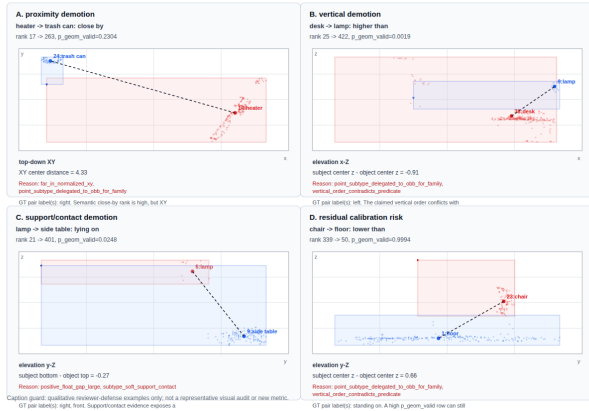


Figure 3: **Geometry-backed qualitative failures.** Open3DSG examples show proximity, relative-vertical, and support/contact cases where semantic plausibility conflicts with object-pair geometry. The rightmost case illustrates residual calibration risk.

after excluding run metadata, although the process-level teardown exit-137 caveat remains.

Failure analysis explains why the Open3DSG metrics change. The full-validation branch produces 82,155 diagnostic rows with zero validation errors; in the 36-case queue, 25 cases are demoted by geometry-aware re-ranking. Figure 3 shows distance, vertical-order, and support-plane contradictions that make symbolic edges unreliable for downstream reasoning.

The same analysis exposes residual risk: ten of 36 sampled rule-violated cases still have $p_{\text{geom_valid}} > 0.9$. Thus $p_{\text{geom_valid}}$ is not a hard validity label, and the paper reports both soft fusion instantiations alongside pooled, RRF, and rule-verified variants.

Limitations

The claim covers contact/support, proximity, and vertical ordering, where reconstructed geometry supports falsifiable checks. Functional and pragmatic predicates remain outside scope. Front/behind is frame-ambiguous, and a left/right variant fails its train/dev strict gate; these are reported as negative scope evidence. Exact-label GT-object evaluation also makes Open3DSG relation-reliability evidence rather than full open-vocabulary generation.

Open3DSG uses a selected non-averaged checkpoint and documented 548/548 recovery branch, not an official leaderboard reproduction. The 533/548 route and historical 377/388–388/388 comparison are sensitivity evidence; broad SOTA and arbitrary-source generality remain blocked.

V is a frozen-verifier measurement. The visual checks are not large-scale human evaluation, and a 488-item protocol still awaits two independent label sets and blinded adjudication; same-agent Codex review is not human evidence. Fusion is also source-dependent: product is stronger on Open3DSG, while rank-based fusion is competitive at VL-SAT and fresh SGFN $K = 100$. The prospective result supports the framework, not formula dominance. support_contact verifier V regresses under product despite aggregate improvement, so the claim is not family-uniform.

SGFN preserves two pre-inference errata for split identity and the full_1160 URL. Eleven self-supported by GT rows lack model edges; no predictions are synthesized, and all remain in the 3,972-row recall denominator. Finally, $p_{\text{geom_valid}}$ and $p_{\text{geom_valid,family}}$ are calibrated reliability scores, not proofs. Residual violations motivate larger human audits, broader physical relations, modern VLM sources, and downstream embodied tests.

Conclusion

This paper studies a scoped reliability failure in 3D Scene Graph relation prediction: semantic relation confidence can rank plausible predicates that are physically inconsistent for the same object pair. We address this failure with a GeoCalib framework that standardizes relation rows, preserves subject-object identity, joins explicit 3D evidence, estimates $p_{\text{geom_valid}}$, and reports recall and geometric violation together. Calibrated product and scale-robust rank-average provide two soft fusion instantiations rather than a claim that one formula is universally optimal.

On Open3DSG as the main open-vocabulary relation-source case study and VL-SAT as a controlled anchor, geometry-consistency reduces violations under measurable recall tradeoffs. On a prospectively frozen SGFN target, both pre-specified soft instantiations pass the aggregate joint gate. GT/counterfactual checks, identity controls, and qualitative failures support the central claim: the contribution is calibrated, identity-preserving, and reportable relation-level physical consistency. Human validity remains pending.

References

- Armeni, I.; He, Z.-Y.; Gwak, J.; Zamir, A. R.; Fischer, M.; Malik, J.; and Savarese, S. 2019. 3D Scene Graph: A Structure for Unified Semantics, 3D Space, and Camera. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 5664–5673.
- Chen, L.; Wang, X.; Lu, J.; Lin, S.; Wang, C.; and He, G. 2024. CLIP-Driven Open-Vocabulary 3D Scene Graph Generation via Cross-Modality Contrastive Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 27863–27873.
- Feng, M.; Hou, H.; Zhang, L.; Wu, Z.; Guo, Y.; and Mian, A. 2023. 3D Spatial Multimodal Knowledge Accumulation for Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9182–9191.
- Geifman, Y.; and El-Yaniv, R. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, volume 30.
- Gu, Q.; Kuwajerwala, A.; Morin, S.; Jatavallabhula, K. M.; Sen, B.; Agarwal, A.; Rivera, C.; Paul, W.; Ellis, K.; Chellappa, R.; Gan, C.; de Melo, C. M.; Tenenbaum, J. B.; Torralba, A.; Shkurti, F.; and Paull, L. 2024. ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 5021–5028.
- Guo, C.; Pleiss, G.; Sun, Y.; and Weinberger, K. Q. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, 1321–1330.
- Heo, K.; Kim, G.; Kim, S.; and Cho, M. 2025. Object-Centric Representation Learning for Enhanced 3D Semantic Scene Graph Prediction. In *Advances in Neural Information Processing Systems*.
- Hou, H.-Y.; Lee, C.-Y.; Sonogashira, M.; and Kawanishi, Y. 2025. FROSS: Faster-Than-Real-Time Online 3D Semantic Scene Graph Generation from RGB-D Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 28818–28827.
- Huang, I.; Bao, Y.; Truong, K.; Zhou, H.; Schmid, C.; Guibas, L.; and Fathi, A. 2025. FirePlace: Geometric Refinements of LLM Common Sense Reasoning for 3D Object Placement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 13466–13476.
- Hughes, N.; Chang, Y.; and Carlone, L. 2022. Hydra: A Real-Time Spatial Perception System for 3D Scene Graph Construction and Optimization. In *Robotics: Science and Systems (RSS)*.
- Koch, S.; Hermosilla, P.; Vaskevicius, N.; Colosi, M.; and Ropinski, T. 2024a. SGRec3D: Self-Supervised 3D Scene Graph Learning via Object-Level Scene Reconstruction. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 3404–3414.
- Koch, S.; Vaskevicius, N.; Colosi, M.; Hermosilla, P.; and Ropinski, T. 2024b. Open3DSG: Open-Vocabulary 3D Scene Graphs from Point Clouds with Queryable Objects and Open-Set Relationships. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14183–14193.
- Liu, Y.; Mei, H.; Zhan, D.; Zhao, J.; Zhou, D.; Dong, B.; and Yang, X. 2026. View-on-Graph: Zero-Shot 3D Visual Grounding via Vision-Language Reasoning on Scene Graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 7386–7394.
- Ma, Y.; Liu, H.; Guo, Y.; Gevers, T.; and Oswald, M. R. 2026. Edge-Centric Relational Reasoning for 3D Scene Graph Prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, 7847–7855.
- Madhavaram, V.; Sengar, V.; De, A.; and Sharma, C. 2026. VIZOR: Viewpoint-Invariant Zero-Shot Scene Graph Generation for 3D Scene Reasoning. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 8584–8595.
- Maggio, D.; Chang, Y.; Hughes, N.; Trang, M.; Griffith, J. D.; Dougherty, C.; Cristofalo, E.; Schmid, L.; and Carlone, L. 2024. Clio: Real-Time Task-Driven Open-Set 3D Scene Graphs. *IEEE Robotics and Automation Letters*, 9(10): 8921–8928.
- Nguyen, M. A.; Tran, Q. H.; Le, B. N.; Pham, T. K.; and Guang, S. Y. 2026. RelWitness: Open-Vocabulary 3D Scene Graph Generation with Visual-Geometric Relation Witnesses. arXiv:2605.20823.
- Ogunleye, M. A.; Abdelrahman, E.; and Lourentzou, I. 2026. 3D-VCD: Hallucination Mitigation in 3D-LLM Embodied Agents through Visual Contrastive Decoding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 40197–40207.
- Peng, S.; Genova, K.; Jiang, C. M.; Tagliasacchi, A.; Pollefeys, M.; and Funkhouser, T. 2023. OpenScene: 3D Scene Understanding With Open Vocabularies. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 815–824.
- Sarkar, S. D.; Miksik, O.; Pollefeys, M.; Barath, D.; and Armeni, I. 2023. SGAligner: 3D Scene Alignment with Scene Graphs. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 21927–21937.
- Saxena, P.; and Chiun, J. 2025. ZING-3D: Zero-shot Incremental 3D Scene Graphs via Vision-Language Models. arXiv:2510.21069.
- Shao, Y.; Zhai, W.; Yang, Y.; Luo, H.; Cao, Y.; and Zha, Z.-J. 2025. GREAT: Geometry-Intention Collaborative Inference for Open-Vocabulary 3D Object Affordance Grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 17326–17336.
- Takmaz, A.; Fedele, E.; Sumner, R. W.; Pollefeys, M.; Tombari, F.; and Engelmann, F. 2023. OpenMask3D: Open-Vocabulary 3D Instance Segmentation. In *Advances in Neural Information Processing Systems*, volume 36, 68367–68390.

- Wald, J.; Dharm, H.; Navab, N.; and Tombari, F. 2020. Learning 3D Semantic Scene Graphs From 3D Indoor Reconstructions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 3961–3970.
- Wang, Z.; Cheng, B.; Zhao, L.; Xu, D.; Tang, Y.; and Sheng, L. 2023. VL-SAT: Visual-Linguistic Semantics Assisted Training for 3D Semantic Scene Graph Prediction in Point Cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 21560–21569.
- Wang, Z.; Su, Y.; Li, C.; Wang, D.; Huang, Y.; Li, X.; and Zhao, B. 2025. Open-Vocabulary Octree-Graph for 3D Scene Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 7037–7047.
- Werby, A.; Huang, C.; Büchner, M.; Valada, A.; and Burgard, W. 2024. Hierarchical Open-Vocabulary 3D Scene Graphs for Language-Grounded Robot Navigation. In *Robotics: Science and Systems (RSS)*.
- Wu, S.-C.; Wald, J.; Tateno, K.; Navab, N.; and Tombari, F. 2021. SceneGraphFusion: Incremental 3D Scene Graph Prediction From RGB-D Sequences. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 7515–7525.
- Xie, Y.; Pagani, A.; and Stricker, D. 2024. SG-PGM: Partial Graph Matching Network with Semantic Geometric Fusion for 3D Scene Graph Alignment and Its Downstream Tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 28401–28411.
- Yeo, Q. X.; Li, Y.; and Lee, G. H. 2025. Statistical Confidence Rescoring for Robust 3D Scene Graph Generation from Multi-View Images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 24999–25008.
- Yu, F.; Deng, Q.; Tang, S.; Li, Y.; and Cheng, L. 2025. Open-World 3D Scene Graph Generation for Retrieval-Augmented Reasoning. Accepted by AAAI 2026, arXiv:2511.05894.
- Zemskova, T.; and Yudin, D. 2025. 3DGraphLLM: Combining Semantic Graphs and Large Language Models for 3D Scene Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 8885–8895.
- Zhang, C.; Delitzas, A.; Wang, F.; Zhang, R.; Ji, X.; Pollefeys, M.; and Engelmann, F. 2025. Open-Vocabulary Functional 3D Scene Graphs for Real-World Indoor Spaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 19401–19413.
- Zhang, C.; Yu, J.; Song, Y.; and Cai, W. 2021. Exploiting Edge-Oriented Reasoning for 3D Point-Based Scene Graph Analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9705–9715.

Reproducibility Checklist

This checklist follows the AAAI-26 reproducibility checklist route checked on 2026-05-26. It is included after the references and is not part of the technical-content page count. Answers refer to the main paper and the Docker-based experiment artifact described in the paper.

General

- Conceptual outline and method description: yes. The paper describes the calibrated geometry-consistency framework, row contract, geometry join, operating points, and recall-violation protocol in Sections 3–4 and Figure 1.
- Separation between facts, claims, and speculation: yes. The paper states the scoped reliability claim, source-specific boundaries, blocked extensions, and limitations explicitly.
- Pedagogical and background references: yes. Related Work cites 3D scene graph prediction, open-vocabulary 3D scene graph methods, geometry-aware relation modeling, and reliability/calibration work.

Theoretical Contributions

- Does this paper make theoretical contributions? no.
- Formal assumptions, theorem statements, proofs, proof sketches, and theoretical-tool citations: NA.
- Empirical demonstrations of theoretical claims and code for theoretical counterexamples: NA.

Datasets

- Does this paper rely on one or more datasets? yes.
- Motivation for selected datasets: yes. Open3DSG is used as the main open-vocabulary relation-source case study, while VL-SAT is used as a controlled reproduced anchor under the same row contract and metrics.
- Novel datasets introduced by this paper: no.
- Novel dataset release and license: NA.
- Citations for existing datasets and sources: yes. The paper cites the relevant 3DSSG, VL-SAT, and Open3DSG sources.
- Public availability of existing datasets: partial. The method uses public research datasets and code routes, but large raw data, meshes, features, checkpoints, and model caches are subject to their original access terms and are not redistributed in the paper PDF.
- Non-public dataset description: NA; no new private dataset is introduced.

Computational Experiments

- Does this paper include computational experiments? yes.
- Development-time hyperparameter search and selection criteria: partial. The paper states the reported operating points, metric scope, and Open3DSG checkpoint selection by train-dev validation loss; exhaustive hyperparameter sweeps are not claimed.

- Preprocessing code: partial. The Docker experiment root records the preprocessing, feature, adapter, geometry-join, metric, table, and failure-analysis commands; large runtime artifacts are rebuilt or mounted rather than embedded in the PDF.
- Source code for experiments and analysis: partial. The tracked repository separates executable code under `src/`, Docker entry points under `configs/`, shell wrappers under `scripts/`, and compact outputs under `results/`; large datasets and model caches are intentionally external.
- Public code release: yes, planned upon publication, subject to anonymization during review and third-party dataset/model licenses.
- Implementation comments: partial. The experiment code and runbooks document the row contract, source adapters, geometry scoring, and metric generation, but final code-release cleanup is still required.
- Randomness and seeds: partial. The main evaluation pipeline is deterministic after fixed predictions, checkpoints, and manifests; full training randomness for reproduced source models is documented as run provenance rather than a multi-seed study.
- Computing infrastructure: partial. The Docker image, compose route, mounted paths, logs, and resource checks are recorded; final submission should include the exact GPU/CPU model and software versions used for the released artifact bundle.
- Evaluation metrics: yes. The paper defines exact-label $R@K$ and Violation@K on the fixed $K = \{5, 10, 20, 50, 100\}$ grid, verifier GT/counterfactual metrics, controls, and failure-analysis criteria.
- Number of runs: yes. Reported source-result tables use one fixed reproduced source run per source/condition under locked manifests.
- Variation or distributional information beyond averages: partial. The paper reports source/family operating points, counterfactual verifier metrics, controls, structured audit, and failure rows, but it does not yet report repeated-run variance.
- Statistical significance tests: no. The current draft reports deterministic held-out recall and violation tradeoffs; no paired significance test is included.
- Final hyperparameters and operating points: partial. The paper and artifact reports list the final operating points, checkpoint choice, denominator policy, and source caveats. The paper-facing Open3DSG route is the 548/548 recovery-policy branch with `OPEN3DSG_MIN_VISIBLE_OBJECTS=2` and relaxed view regeneration for two scans; the exact-label measured-family denominator is 3,972, and residual calibration risk is reported through the failure-analysis artifacts. A final release table should collect every threshold and configuration in one place.